AERIS Expert Review Packet

Anomaly

Source / metric	tceq / Carbon monoxide (co)
Observed / expected baseline	0.9 / 0.353571 ppm
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-06-01 11:00 UTC
Location	29.8144, -95.3878
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	9 / 407	43.99 to 100; mean 78.6385	100 at 2026-06-01 10:55Z; 5 min before event
asos	Air temperature (temperature)	degC	9 / 407	21.6667 to 34; mean 27.9773	25 at 2026-06-01 10:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	9 / 411	0 to 360; mean 94.9878	160 at 2026-06-01 10:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	9 / 411	0 to 7.71666; mean 2.0415	2.05778 at 2026-06-01 10:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	11 / 132	5878.05 to 5901.83; mean 5890.49	5884.74 at 2026-06-01 12:00Z; 1 h after event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	11 / 132	14.9006 to 2020.13; mean 639.555	132.33 at 2026-06-01 12:00Z; 1 h after event
noaa_gfs	Precipitable water (precipitable_water)	mm	11 / 132	38.3674 to 53.6474; mean 46.904	45.9772 at 2026-06-01 12:00Z; 1 h after event
noaa_gfs	Surface pressure (surface_pressure)	hPa	11 / 132	1002.93 to 1013.92; mean 1009.88	1010.74 at 2026-06-01 12:00Z; 1 h after event
noaa_gfs	850-hPa temperature (t_850)	degC	11 / 132	16.5256 to 20.7457; mean 18.2336	17.8526 at 2026-06-01 12:00Z; 1 h after event
noaa_gfs	10-m eastward wind (u_10m)	m/s	11 / 132	-3.82952 to 1.08048; mean -1.2755	-0.453428 at 2026-06-01 12:00Z; 1 h after event
noaa_gfs	10-m northward wind (v_10m)	m/s	11 / 132	-1.54819 to 4.0161; mean 1.4832	1.04485 at 2026-06-01 12:00Z; 1 h after event
openaq	Ozone (ozone)	ppm	13 / 535	-0.002 to 0.094; mean 0.0206	0.008 at 2026-06-01 11:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	2 / 84	12 to 110; mean 30.5238	26 at 2026-06-01 11:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	28 / 2018	-4 to 34.5; mean 5.8885	12.7 at 2026-06-01 11:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 360	0 to 100; mean 41.7694	40 at 2026-06-01 10:57Z; 2 min before event
openweather	Relative humidity (humidity)	percent	5 / 360	57 to 95; mean 78.1528	95 at 2026-06-01 10:57Z; 2 min before event
openweather	Precipitation rate (precipitation)	mm/h	5 / 360	0 to 6.48; mean 0.06	0 at 2026-06-01 10:57Z; 2 min before event
openweather	Surface pressure (pressure)	hPa	5 / 360	1010 to 1015; mean 1013.03	1014 at 2026-06-01 10:57Z; 2 min before event
openweather	Air temperature (temperature)	degC	5 / 360	23.96 to 34.52; mean 28.6035	24.3 at 2026-06-01 10:57Z; 2 min before event
openweather	Wind direction (wind_direction)	degree	5 / 360	0 to 360; mean 130.339	0 at 2026-06-01 10:57Z; 2 min before event
openweather	Wind speed (wind_speed)	m/s	5 / 360	0 to 7.2; mean 2.5835	0 at 2026-06-01 10:57Z; 2 min before event
purpleair	PM2.5 (pm25)	ug/m3	65 / 3103	0 to 58; mean 6.1513	8.3 at 2026-06-01 11:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; 8 h 42 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; 8 h 42 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	4 / 4	0.028772 to 0.0302015; mean 0.0295	0.0300364 at 2026-06-01 19:42Z; 8 h 42 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; 8 h 42 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	5 / 5	0.000117197 to 0.000169173; mean 0.0001	0.000166437 at 2026-06-01 19:42Z; 8 h 42 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; 8 h 42 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	6 / 6	1.483e-05 to 4.38402e-05; mean 0	3.77616e-05 at 2026-06-01 19:42Z; 8 h 42 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; 8 h 42 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; 8 h 42 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	5 / 5	-9.16616e-06 to 0.000116468; mean 0	1.74952e-05 at 2026-06-01 19:42Z; 8 h 42 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	8 / 8	1 to 1; mean 1	1 at 2026-06-01 19:42Z; 8 h 42 min after event
tceq	Carbon monoxide (co)	ppm	3 / 194	0.1 to 0.9; mean 0.2216	0.9 at 2026-06-01 11:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	13 / 805	-2.6 to 34.4; mean 4.8637	10.1 at 2026-06-01 11:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 191	-0.5 to 1.6; mean -0.0696	-0.2 at 2026-06-01 11:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	06-01 00Z 82.65, 06Z 95.36, 12Z 70.61, 18Z 60.8, 06-02 00Z 84.01, 06Z 94.18, 12Z 76.34, 18Z 65.27
asos	Air temperature (temperature)	06-01 00Z 26.77, 06Z 24.39, 12Z 29.37, 18Z 31.67, 06-02 00Z 27.27, 06Z 25.14, 12Z 28.78, 18Z 30.34
asos	Wind direction (wind_direction)	06-01 00Z 133.9, 06Z 21.32, 12Z 80.57, 18Z 135.6, 06-02 00Z 129.4, 06Z 52.16, 12Z 98.15, 18Z 112.7
asos	Wind speed (wind_speed)	06-01 00Z 2.541, 06Z 0.3106, 12Z 1.281, 18Z 3.782, 06-02 00Z 1.998, 06Z 0.5851, 12Z 1.982, 18Z 4.116
noaa_gfs	500-hPa geopotential height (gh_500)	05-31 00Z 5882, 06Z 5891, 12Z 5880, 18Z 5896, 06-01 00Z 5891, 06Z 5900, 12Z 5885, 18Z 5900, 06-02 00Z 5888, 06Z 5898, 12Z 5881, 18Z 5894
noaa_gfs	Planetary boundary layer height (pbl_height)	05-31 00Z 611.8, 06Z 256.5, 12Z 148.8, 18Z 1584, 06-01 00Z 732.2, 06Z 225.8, 12Z 67.97, 18Z 1700, 06-02 00Z 564.7, 06Z 86.98, 12Z 42.25, 18Z 1654
noaa_gfs	Precipitable water (precipitable_water)	05-31 00Z 44.02, 06Z 40.24, 12Z 43.83, 18Z 45.68, 06-01 00Z 47.21, 06Z 46.22, 12Z 46.21, 18Z 48.98, 06-02 00Z 51.86, 06Z 49.79, 12Z 50.16, 18Z 48.67
noaa_gfs	Surface pressure (surface_pressure)	05-31 00Z 1007, 06Z 1008, 12Z 1010, 18Z 1010, 06-01 00Z 1008, 06Z 1011, 12Z 1011, 18Z 1011, 06-02 00Z 1009, 06Z 1011, 12Z 1011, 18Z 1012
noaa_gfs	850-hPa temperature (t_850)	05-31 00Z 18.85, 06Z 20.15, 12Z 18.83, 18Z 16.91, 06-01 00Z 18.45, 06Z 18.58, 12Z 17.84, 18Z 17.33, 06-02 00Z 18.76, 06Z 18.54, 12Z 17.59, 18Z 16.97
noaa_gfs	10-m eastward wind (u_10m)	05-31 00Z -2.182, 06Z -0.9521, 12Z -0.6771, 18Z -1.243, 06-01 00Z -2.565, 06Z -0.7446, 12Z -0.4489, 18Z -1.458, 06-02 00Z -2.194, 06Z -0.3987, 12Z -0.483, 18Z -1.96
noaa_gfs	10-m northward wind (v_10m)	05-31 00Z 3.1, 06Z 2.54, 12Z 1.595, 18Z 2.177, 06-01 00Z 2.902, 06Z 1.967, 12Z 0.7749, 18Z 0.7166, 06-02 00Z 2.65, 06Z 1.596, 12Z -1.005, 18Z -1.215
openaq	Ozone (ozone)	06-01 06Z 0.009731, 12Z 0.01646, 18Z 0.04762, 06-02 00Z 0.01781, 06Z 0.007731, 12Z 0.01115, 18Z 0.03371
openaq	PM10 (pm10)	06-01 06Z 31.42, 12Z 44.58, 18Z 30.83, 06-02 00Z 25.17, 06Z 22.08, 12Z 31.17, 18Z 28.42
openaq	PM2.5 (pm25)	06-01 06Z 6.451, 12Z 5.868, 18Z 6.815, 06-02 00Z 5.303, 06Z 4.973, 12Z 6.72, 18Z 5.216
openweather	Cloud cover (cloud_cover)	05-30 18Z 47, 05-31 00Z 22.33, 06Z 31.73, 12Z 61.93, 18Z 55.3, 06-01 00Z 20.7, 06Z 41.5, 12Z 43.47, 18Z 44.27, 06-02 00Z 20.8, 06Z 24.37, 12Z 54.9, 18Z 86.52
openweather	Relative humidity (humidity)	05-30 18Z 66, 05-31 00Z 78.43, 06Z 89.6, 12Z 74.57, 18Z 64.47, 06-01 00Z 81.03, 06Z 92.27, 12Z 74.93, 18Z 64.9, 06-02 00Z 80, 06Z 91.3, 12Z 78.7, 18Z 67.96
openweather	Precipitation rate (precipitation)	05-30 18Z 0, 05-31 00Z 0, 06Z 0, 12Z 0, 18Z 0, 06-01 00Z 0, 06Z 0.005, 12Z 0, 18Z 0.03, 06-02 00Z 0, 06Z 0, 12Z 0, 18Z 0.8216
openweather	Surface pressure (pressure)	05-30 18Z 1010, 05-31 00Z 1011, 06Z 1012, 12Z 1014, 18Z 1011, 06-01 00Z 1013, 06Z 1014, 12Z 1014, 18Z 1012, 06-02 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1013
openweather	Air temperature (temperature)	05-30 18Z 31.44, 05-31 00Z 27.36, 06Z 24.86, 12Z 29.25, 18Z 32.05, 06-01 00Z 27.42, 06Z 24.92, 12Z 29.78, 18Z 32.53, 06-02 00Z 28.22, 06Z 25.73, 12Z 29.28, 18Z 31.91
openweather	Wind direction (wind_direction)	05-30 18Z 154, 05-31 00Z 165, 06Z 125.2, 12Z 157.6, 18Z 142.1, 06-01 00Z 151, 06Z 79.53, 12Z 122.5, 18Z 133.8, 06-02 00Z 166.2, 06Z 68.2, 12Z 121.1, 18Z 127.3
openweather	Wind speed (wind_speed)	05-30 18Z 4.938, 05-31 00Z 3.893, 06Z 1.936, 12Z 2.946, 18Z 4.11, 06-01 00Z 3.397, 06Z 1.041, 12Z 1.079, 18Z 3.923, 06-02 00Z 3.192, 06Z 0.8987, 12Z 1.722, 18Z 2.45
purpleair	PM2.5 (pm25)	06-01 00Z 5.865, 06Z 7.763, 12Z 5.952, 18Z 6.977, 06-02 00Z 5.427, 06Z 5.803, 12Z 6.927, 18Z 4.514

Source	Metric	Values
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	CO column (s5p_co_column)	06-01 18Z 0.03012, 06-02 18Z 0.02882
sentinel5p	CO granules available (s5p_co_granule_available)	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	05-31 18Z 0.0001186, 06-01 18Z 0.0001645, 06-02 18Z 0.0001692
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	05-31 18Z 1.667e-05, 06-01 18Z 4.08e-05, 06-02 18Z 4.07e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	05-31 18Z 0.0001165, 06-01 18Z 1.316e-05, 06-02 18Z -8.155e-06
sentinel5p	SO2 granules available (s5p_so2_granule_available)	05-31 18Z 1, 06-01 18Z 1, 06-02 18Z 1
tceq	Carbon monoxide (co)	05-31 06Z 0.1833, 12Z 0.2167, 18Z 0.22, 06-01 00Z 0.25, 06Z 0.2833, 12Z 0.2529, 18Z 0.2278, 06-02 00Z 0.2, 06Z 0.2333, 12Z 0.2056, 18Z 0.1667
tceq	Nitrogen dioxide (no2)	05-31 06Z 3.377, 12Z 3.055, 18Z 2.811, 06-01 00Z 3.459, 06Z 5.427, 12Z 6.123, 18Z 5.614, 06-02 00Z 4.532, 06Z 5.082, 12Z 7.521, 18Z 6.109
tceq	Sulfur dioxide (so2)	05-31 06Z -0.2188, 12Z -0.2467, 18Z -0.1556, 06-01 00Z -0.1556, 06Z -0.1778, 12Z 0.1889, 18Z 0.08333, 06-02 00Z -0.1444, 06Z -0.1294, 12Z -0.01176, 18Z 0.1556

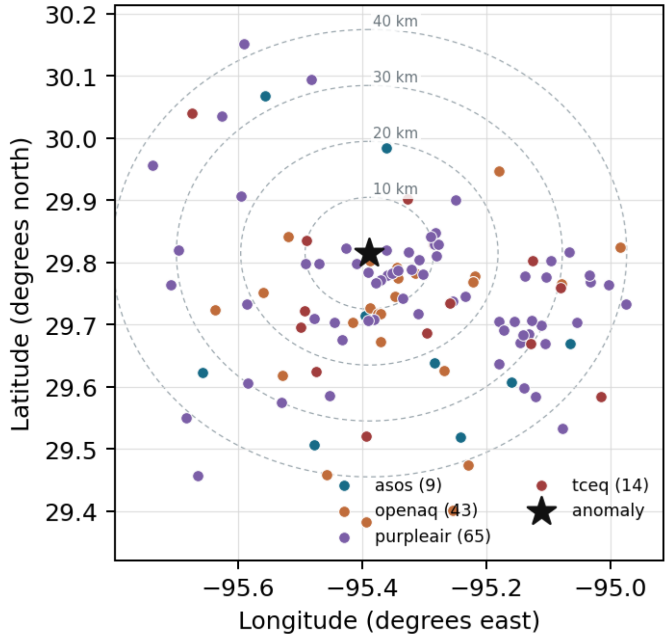
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

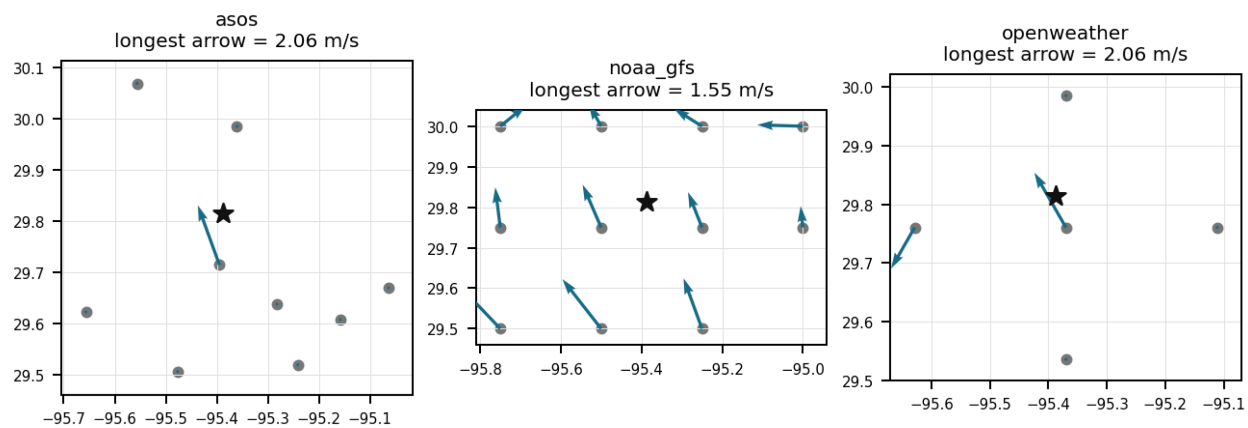
Source	Metric	Values
asos	Relative humidity (humidity)	MCJ (11.184 km) 76.5, IAH (19.084 km) 78.65, HOU (22.147 km) 76.77, EFD (31.936 km) 78.31, DWH (32.534 km) 83.57, SGR (33.607 km) 79.28, T41 (35.177 km) 80.68, AXH (35.344 km) 75.89, +1 more
asos	Air temperature (temperature)	MCJ (11.184 km) 28.57, IAH (19.084 km) 28.12, HOU (22.147 km) 28.42, EFD (31.936 km) 28.23, DWH (32.534 km) 26.77, SGR (33.607 km) 28.17, T41 (35.177 km) 27.8, AXH (35.344 km) 27.87, +1 more
asos	Wind direction (wind_direction)	MCJ (11.184 km) 110.7, IAH (19.084 km) 103, HOU (22.147 km) 114.9, EFD (31.936 km) 94.65, DWH (32.534 km) 75.96, SGR (33.607 km) 109.8, T41 (35.177 km) 110.6, AXH (35.344 km) 64.47, +1 more
asos	Wind speed (wind_speed)	MCJ (11.184 km) 3.285, IAH (19.084 km) 1.926, HOU (22.147 km) 2.386, EFD (31.936 km) 2.249, DWH (32.534 km) 0.9851, SGR (33.607 km) 2.401, T41 (35.177 km) 2.32, AXH (35.344 km) 1.226, +1 more
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.50 (12.98 km) 5890, gfs:29.75,-95.25 (15.105 km) 5891, gfs:30.00,-95.50 (23.3 km) 5890, gfs:30.00,-95.25 (24.544 km) 5890, gfs:29.75,-95.75 (35.679 km) 5890, gfs:29.50,-95.50 (36.601 km) 5891, gfs:29.50,-95.25 (37.409 km) 5892, gfs:29.75,-95.00 (38.106 km) 5891, +3 more
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.50 (12.98 km) 815.9, gfs:29.75,-95.25 (15.105 km) 784.8, gfs:30.00,-95.50 (23.3 km) 933.2, gfs:30.00,-95.25 (24.544 km) 736.9, gfs:29.75,-95.75 (35.679 km) 573.7, gfs:29.50,-95.50 (36.601 km) 566.1, gfs:29.50,-95.25 (37.409 km) 546.2, gfs:29.75,-95.00 (38.106 km) 555.7, +3 more

Source	Metric	Values
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.50 (12.98 km) 46.72, gfs:29.75,-95.25 (15.105 km) 47.24, gfs:30.00,-95.50 (23.3 km) 46.46, gfs:30.00,-95.25 (24.544 km) 47.06, gfs:29.75,-95.75 (35.679 km) 46.4, gfs:29.50,-95.50 (36.601 km) 46.86, gfs:29.50,-95.25 (37.409 km) 47.4, gfs:29.75,-95.00 (38.106 km) 47.75, +3 more
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.50 (12.98 km) 1010, gfs:29.75,-95.25 (15.105 km) 1011, gfs:30.00,-95.50 (23.3 km) 1008, gfs:30.00,-95.25 (24.544 km) 1010, gfs:29.75,-95.75 (35.679 km) 1009, gfs:29.50,-95.50 (36.601 km) 1011, gfs:29.50,-95.25 (37.409 km) 1012, gfs:29.75,-95.00 (38.106 km) 1012, +3 more
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.50 (12.98 km) 18.1, gfs:29.75,-95.25 (15.105 km) 18.28, gfs:30.00,-95.50 (23.3 km) 18.14, gfs:30.00,-95.25 (24.544 km) 18.27, gfs:29.75,-95.75 (35.679 km) 18.12, gfs:29.50,-95.50 (36.601 km) 18.35, gfs:29.50,-95.25 (37.409 km) 18.36, gfs:29.75,-95.00 (38.106 km) 18.34, +3 more
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.50 (12.98 km) -1.078, gfs:29.75,-95.25 (15.105 km) -1.439, gfs:30.00,-95.50 (23.3 km) -1.02, gfs:30.00,-95.25 (24.544 km) -1.673, gfs:29.75,-95.75 (35.679 km) -0.9367, gfs:29.50,-95.50 (36.601 km) -1.709, gfs:29.50,-95.25 (37.409 km) -1.63, gfs:29.75,-95.00 (38.106 km) -1.455, +3 more
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.50 (12.98 km) 1.723, gfs:29.75,-95.25 (15.105 km) 1.451, gfs:30.00,-95.50 (23.3 km) 1.258, gfs:30.00,-95.25 (24.544 km) 1.248, gfs:29.75,-95.75 (35.679 km) 1.532, gfs:29.50,-95.50 (36.601 km) 1.503, gfs:29.50,-95.25 (37.409 km) 1.663, gfs:29.75,-95.00 (38.106 km) 1.652, +3 more
openaq	Ozone (ozone)	319 (10.025 km) 0.02202, 2046 (10.112 km) 0.01824, 310 (11.331 km) 0.02576, 3378 (15.437 km) 0.01724, 325 (16.833 km) 0.01859, 2039 (16.937 km) 0.02192, 311 (17.015 km) 0.02202, 320 (24.007 km) 0.01546, +5 more
openaq	PM10 (pm10)	13380082 (15.437 km) 38.88, 271 (29.74 km) 22.17
openaq	PM2.5 (pm25)	2071327 (0.02 km) 10.82, 10107877 (1.373 km) 6.46, 8967193 (4.942 km) 6.691, 15596557 (6.193 km) 5.81, 8967217 (7.937 km) 5.622, 15666344 (8.753 km) 2.571, 12286630 (9.769 km) 2.06, 2071331 (10.112 km) 12.08, +20 more
openweather	Cloud cover (cloud_cover)	grid:center (6.25 km) 41.92, grid:north (19.05 km) 56.43, grid:west (23.99 km) 43.18, grid:east (27.371 km) 38.85, grid:south (31.027 km) 28.47
openweather	Relative humidity (humidity)	grid:center (6.25 km) 77.38, grid:north (19.05 km) 77.61, grid:west (23.99 km) 77.31, grid:east (27.371 km) 81.86, grid:south (31.027 km) 76.61
openweather	Precipitation rate (precipitation)	grid:center (6.25 km) 0.06528, grid:north (19.05 km) 0.126, grid:west (23.99 km) 0.1036, grid:east (27.371 km) 0.003333, grid:south (31.027 km) 0.001667
openweather	Surface pressure (pressure)	grid:center (6.25 km) 1013, grid:north (19.05 km) 1013, grid:west (23.99 km) 1013, grid:east (27.371 km) 1013, grid:south (31.027 km) 1013
openweather	Air temperature (temperature)	grid:center (6.25 km) 29.01, grid:north (19.05 km) 28.69, grid:west (23.99 km) 28.78, grid:east (27.371 km) 28.2, grid:south (31.027 km) 28.35
openweather	Wind direction (wind_direction)	grid:center (6.25 km) 135.2, grid:north (19.05 km) 126.4, grid:west (23.99 km) 152.6, grid:east (27.371 km) 108.7, grid:south (31.027 km) 128.7
openweather	Wind speed (wind_speed)	grid:center (6.25 km) 3.377, grid:north (19.05 km) 2.463, grid:west (23.99 km) 2.217, grid:east (27.371 km) 2.432, grid:south (31.027 km) 2.428
purpleair	PM2.5 (pm25)	223813 (2.634 km) 6.127, 246011 (2.666 km) 4.948, 302037 (3.437 km) 7.242, 155367 (3.772 km) 6.735, 127451 (4.843 km) 1.474, 186303 (5.079 km) 6.65, 27821 (5.147 km) 0.3042, 296101 (5.382 km) 6.562, +57 more
tceq	Carbon monoxide (co)	482011052 (0.0 km) 0.3508, 482011035 (15.44 km) 0.1688, 482011039 (29.756 km) 0.1446
tceq	Nitrogen dioxide (no2)	482011052 (0.0 km) 9.752, 482010047 (10.018 km) 5.362, 482010024 (11.323 km) 0.1045, 482011066 (14.45 km) 4.466, 482011035 (15.44 km) 7.861, 482010416 (16.841 km) 4.753, 482010055 (17.021 km) 3.105, 482010026 (25.342 km) 8.411, +5 more
tceq	Sulfur dioxide (so2)	482011035 (15.44 km) -0.06061, 482010051 (22.762 km) -0.3062, 482011039 (29.756 km) 0.1767

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide
Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.
Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.
Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.
Missing or thin evidence means Unsure, not Invalid.
A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.
Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 2 of 36

The Houston tceq monitor recorded a severe local carbon monoxide enhancement to 0.9 ppm at 2026-06-01T11:00:00+00:00, while other nearby tceq carbon monoxide monitors in the 50 km domain had much lower period means, which supports a localized source rather than a domain-wide carbon monoxide event.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 3 of 36

The air quality anomaly in Houston, Texas on June 1, 2026 is characterized by elevated levels of PM2.5.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 4 of 36

The air quality anomaly in Houston, Texas is not caused by CO, NO2, or SO2 levels.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 5 of 36

The stagnation-and-trapping hypothesis is strongly supported because near the anomaly time Houston-area surface winds were very weak, local humidity was very high, and modeled planetary boundary layer height was exceptionally low at about 132 m near 12 UTC, all of which are conditions that inhibit dilution of primary pollutants such as carbon monoxide.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 6 of 36

The air quality anomaly in Houston, Texas is caused by a combination of high temperatures and wind direction.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 7 of 36

The temperature inversion caused by the heat dome over Houston led to a stagnation of air, resulting in poor air quality.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 8 of 36

The anomalous pollutant was carbon monoxide at tceq monitor 482011052 located at 29.814, -95.388 in Houston, Texas.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 9 of 36

At the time of the carbon monoxide anomaly on June 1, 2026, at 11:00 UTC, nitrogen dioxide concentrations at TCEQ monitor 482011052 were measured at 10.1 ppb.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 10 of 36

The broad regional or biomass-burning hypothesis is weakened because particulate matter remained modest near the anomaly, ozone was low near the anomaly time, and the later-day satellite carbon monoxide column increase was only slight without corroborating high aerosol loading or sulfur dioxide enhancement, which is not the pattern expected for a major regional smoke or widespread combustion episode.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 11 of 36

Shallow atmospheric boundary layer heights below 100 meters combined with light surface winds under 1.5 m/s restricted vertical and horizontal dispersion around 11:00 UTC on June 1, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 12 of 36

Regional smoke or secondary-photochemical explanations are less supported because particulate matter was not regionally elevated at anomaly time, ozone near the anomaly was low at anomaly time, and later satellite observations did not show a strong coincident sulfur-rich regional plume over the site.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 13 of 36

There is no evidence to suggest that the air quality anomaly in Houston, Texas on June 1, 2026 was caused by any specific weather or environmental factors.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 14 of 36

A severe carbon monoxide anomaly reaching 0.9 ppm was observed at TCEQ monitor 482011052 (29.814, -95.388) on June 1, 2026, at 11:00 UTC against an expected baseline of 0.35 ppm.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 15 of 36

The prevailing wind direction and speed, which were unfavorable for dispersing pollutants, exacerbated the air quality issue.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 16 of 36

The air quality anomaly in Houston, Texas on June 1, 2026 is also characterized by elevated levels of NO_2.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 17 of 36

The Houston anomaly occurred under stagnant early-morning conditions characterized by very light surface winds, very high humidity, and a shallow planetary boundary layer, which would strongly increase near-source carbon monoxide accumulation.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 18 of 36

A major sulfur-rich industrial release or a regional smoke event is a less likely primary cause of the Houston carbon monoxide spike, because sulfur dioxide near the area was not elevated, particulate matter remained modest, and the later satellite overpass showed only modest carbon monoxide column enhancement rather than an extreme regional plume.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 19 of 36

The temperature during the anomaly was around 24.3 degC, which is lower than the mean value of 28.6035 degC.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 20 of 36

Primary combustion emissions from local vehicular traffic or industrial operations accumulated in the surface boundary layer, elevating carbon monoxide and nitrogen dioxide levels at monitoring site 482011052.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 21 of 36

GFS weather model data and ASOS surface observations indicated a shallow boundary layer height under 100 meters and low surface wind speeds during the morning of June 1, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 22 of 36

Sentinel-5P TROPOMI satellite imagery recorded an elevated carbon monoxide column density averaging 0.0301 mol/m^2 over the Houston area during its afternoon pass on June 1, 2026, confirming regional enrichment of atmospheric carbon monoxide.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 23 of 36

A shallow and stagnant early-morning boundary layer likely amplified near-surface carbon monoxide concentrations at the Houston monitor by suppressing vertical dilution, because boundary-layer height was extremely low near the event time and surface winds were very weak with very high humidity.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 24 of 36

Ground-based air quality monitoring from Texas Commission on Environmental Quality and OpenAQ recorded coincident elevated concentrations of carbon monoxide, nitrogen dioxide, and fine particulate matter at the anomaly location at 11:00 UTC on June 1, 2026, pointing to fresh combustion emissions.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 25 of 36

Global Forecast System model data and Automated Surface Observing System measurements indicate that planetary boundary layer heights dropped below 100 meters alongside near-calm surface winds between 06:00 UTC and 12:00 UTC on June 1, 2026, creating stagnant atmospheric conditions that suppressed pollutant dispersion.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 26 of 36

The tceq carbon monoxide measurement at 2026-06-01T11:00:00+00:00 was 0.9 ppm, compared with an expected value of 0.35357142857142854 ppm, giving a z-score of 5.196152422706633 and a severe anomaly classification.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 27 of 36

Satellite measurements detected elevated tropospheric column abundances of nitrogen dioxide and carbon monoxide over the Houston area during the same period.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 28 of 36

The anomaly occurred on June 1st at around 11:00 AM (UTC) and was detected by a PurpleAir sensor located approximately 2.634 km from the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 29 of 36

The high levels of particulate matter (PM2.5) from industrial activities and vehicle emissions in the Houston area contributed to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 30 of 36

The air quality anomaly is related to particulate matter (PM2.5) with a value of 8.3 ug/m3, which is elevated compared to the mean value of 6.1513 ug/m3.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 31 of 36

TCEQ monitoring station 482011052 recorded a carbon monoxide peak of 0.9 ppm alongside elevated nitrogen dioxide levels on June 1, 2026, at 11:00 UTC in Houston.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 32 of 36

Sentinel-5P satellite retrievals detected elevated total carbon monoxide column values reaching approximately 0.030 mol/m^2 over Houston on June 1, 2026.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 33 of 36

The tceq carbon monoxide network within 50 km had a mean concentration of 0.2216 ppm over the context window, while the nearest monitor recorded 0.9 ppm at the anomaly time.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 34 of 36

The local combustion plume hypothesis is supported because the anomalous tceq carbon monoxide value of 0.9 ppm occurred at one monitor while the other tceq carbon monoxide monitors in the 50 km window had much lower mean concentrations, indicating a spatially localized enhancement rather than a network-wide rise.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 35 of 36

At the time of the carbon monoxide anomaly on June 1, 2026, at 11:00 UTC, fine particulate matter (PM2.5) reached 12.7 ug/m3 at OpenAQ sensor 2071327 located 0.02 km from the anomaly.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Claim 36 of 36

A nearby fresh combustion plume from traffic, diesel equipment, marine activity, or industrial combustion is a likely direct cause of the Houston carbon monoxide spike, because carbon monoxide rose sharply at one tceq monitor while co-located nitrogen dioxide was elevated and ozone remained low, which is consistent with a local primary-emission plume rather than aged photochemical pollution.

Mark exactly one:	☐ V	☐ I	☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.